The 1.0–1.2 TeV Thermal Higgsino Against Planck, LZ and XENONnT: Toy-Model Relic and Inelastic-Recoil Calculations with a Literature Survey

HEP-Genesis-Agent session record
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Abstract

We report a two-pass investigation of where the pure higgsino thermal relic, at $m_\chi \approx 1.0$–1.2 TeV, sits relative to the Planck abundance and to current xenon direct-detection limits. We combined calculations from an inelastic-dark-matter (inelastic-DM) tool server with an abstract- and HTML-level literature survey.

The server's relic model is a single-channel Kolb–Turner freeze-out with a built-in higgsino cross section $\langle\sigma v\rangle = 2.5 \times 10^{-26}\,(1100\,\mathrm{GeV}/m)^2\ \mathrm{cm}^3\,\mathrm{s}^{-1}$. It gives $\Omega h^2 = 0.0731$, 0.0898, 0.1083, 0.1285 and 0.1503 at $m = 0.9$–1.3 TeV, and crosses the Planck value $\Omega_c h^2 = 0.120 \pm 0.001$ at $m \simeq 1159$–1160 GeV. That crossing reflects the calibration and is not a Sommerfeld or coannihilation prediction. The cited literature places the thermal mass at about 1.08–1.2 TeV.

In the tool, elastic Z-mediated scattering at zero splitting predicts $\sim 1.1 \times 10^7$ LZ events and is excluded. With the tool's hard halo cutoff, $v_{\mathrm{esc}} + v_E = 776$ km/s, about one event in the LZ 2.84 t·yr extended window occurs only near $\delta \approx 352$ keV, and the rate closes by 370 keV. The literature quotes an IceCube solar-capture bound $\delta > 566$ keV, which lies beyond xenon's inelastic reach.

For elastic spin-independent scattering, the pure higgsino is only bounded, at $\sigma_{\mathrm{SI}} \lesssim 10^{-48}\,\mathrm{cm}^2$. At 1.1 TeV this is at least $\sim 33\times$ below the LZ 4.2 t·yr limit, at least $\sim 400\times$ below the XENONnT 3.1 t·yr limit, and at least $\sim 5.5\times$ below the xenon neutrino floor.

Several tool flags proved unreliable, halo dependence could not be tested, and indirect-detection conclusions rest on abstracts only.

1 Introduction

The user asked four things. First, a survey of recent arXiv work on the higgsino thermal relic target at 1.0–1.2 TeV. Second, use of the available inelastic-DM tools to compute the relic density at several masses. Third, a comparison with LZ/XENONnT sensitivities and the Planck abundance. Fourth, a statement of where the higgsino sits relative to current limits. The same request was issued twice. The first pass ran with a largely non-functional arXiv API layer. The second pass, after the API recovered, added verbatim citations, elastic spin-independent (SI) limits at TeV masses, a relic cross-section sensitivity test, and a gaugino-scale translation of the mass splitting.

The research plan evolved through the following sub-questions:

1. The tool's relic curve and its Planck crossing.
2. The inelastic LZ event counts and the kinematic edge of xenon.
3. Verbatim literature support for the key numbers.
4. The width of the splitting window that gives about one LZ event, and its halo sensitivity.
5. The indirect-detection status of the higgsino, added because the server's annihilation treatment was judged physically wrong.
6. Elastic SI placement relative to limits and the neutrino fog.
7. The gaugino-mass scale implied by the splitting.

Much of the 2026-dated literature concerns a single LZ nuclear-recoil event at 248 keV (arXiv:2609.02823). It enters here only insofar as it bears on the higgsino's position relative to limits.

2 Methods

Tool server. Calculations used an `inelastic-dm-server` with local dispatch. Its self-description cites the Lewin–Smith rate formula, Kolb–Turner freeze-out, Fermi-LAT limits, and Planck 2018 for CMB energy injection. It stores $\Omega_{\rm DM}h^2 = 0.12 \pm 0.0012$, the LZ event at 248 ± 23 keV, and an exposure of 2.84 t·yr. The tools used were:

- `compute_relic_density`, with inputs m_χ, an `is_higgsino` flag and an optional $\langle\sigma v\rangle$;
- `predict_recoil`, with inputs m_χ, δ, σ_n and the higgsino flag. It returns expected counts in the LZ extended nuclear-recoil window, $v_{\rm min}$ at the spectrum peak and at 248 keV, and an accessibility flag;
- `check_all_constraints`, covering relic, LZ, Fermi-LAT and CMB $p_{\rm ann}$;
- `scan_mass_splitting` and `plot_constraint_summary`;
- `predict_annual_modulation` and `plot_recoil_spectrum`.

With the higgsino flag set, the tool assigns $\sigma_n = 7.43 \times 10^{-39}\,{\rm cm}^2$ per nucleon, corresponding to Z exchange. None of the tools accepts halo parameters. The halo is hard-coded with $v_{\rm esc} + v_E = 776$ km/s.

Runs. The relic density was computed at $m = 900$, 1000, 1100, 1200, 1300 and 1159 GeV. Recoil predictions at $m = 1100$ GeV covered $\delta = 0$, 300, 330, 350, 355, 360, 365, 370, 375, 380, 390, 500, 566 and 700 keV. Further recoil runs were made at $m = 1000$ and 1200 GeV ($\delta = 380$ keV), at 1159 GeV ($\delta = 360$ keV), and at 1160 GeV ($\delta = 350$ and 360 keV).

Constraint checks were run at:

- $m = 1100$ GeV with $\delta = 0$, 200, 350, 400, 566 and 1000 keV;
- $m = 1159$ GeV with $\delta = 400$ keV;
- $m = 1080$, 1160 and 1200 GeV with $\delta = 350$ keV.

Two parameter scans were made: 17×17 over 800–1400 GeV $\times$ 0–800 keV, and 15×15 over 1000–1200 GeV $\times$ 0–700 keV. The second wrote to the same output file as the first. To probe the relic normalization, $\langle\sigma v\rangle$ was varied at 1100 GeV (2.0, 2.25, 3.0×10^{-26} cm^3/s) with the higgsino flag off. This was necessary because with the flag on the explicit argument was ignored and the built-in value was echoed back.

Hand calculations. The kinematic edge was estimated by hand from $\delta_{\max} = \mu v_{\max}^2/2$, with $m_{\rm Xe} \approx 122$ GeV and $\mu = 109.8$ GeV at 1.1 TeV. The gaugino scale was estimated from the tree-level relation quoted from arXiv:1410.4549.

Literature access. In the first pass, the direct arXiv API search returned HTTP 503/429 errors. Metadata, download and `cite` calls returned "not found" even for arXiv:1807.06209, and the full-paper downloader failed with `'Search' object has no attribute 'results'`. Only an OpenAlex-backed search returning abstracts worked.

In the second pass, metadata lookups succeeded and `cite` recorded verbatim abstract quotes for five papers. A sub-agent read arXiv HTML pages (downloads still failed), the LZ HEPData record 155182 (table t1), and O'Hare's public neutrino-floor file `Xe_SI.txt`. A `cite` attempt on arXiv:2410.17036 failed its quote check.

Figure extraction from arXiv:2410.17036 and arXiv:2502.18005 returned only event-distribution and calibration plots, not the SI exclusion curves. These figures carry no information used in the analysis and are not shown. The full tool-call log and code are in the supplement.

Reporting discrepancy. In both passes, the assistant's summary text claimed that no inelastic-DM or relic-density tools were available and that no relic densities were computed. The tool record contradicts this: the server tools were called and returned results. The results below are taken from those tool outputs.

3 Results

3.1 Literature survey of the thermal target

All values in this subsection come from abstracts. The 2026 identifiers were first found only through OpenAlex and were later confirmed by arXiv metadata in the second pass, at least for arXiv:2609.02775 and arXiv:2609.02608.

The verbatim-cited mass values are as follows:

- arXiv:2410.15733 states that in the thermal-relic scenario "the observed DM abundance is realized with the Higgsino mass of $\sim$1.2 TeV". The abstract says it includes coannihilation and the Sommerfeld effect. The paper itself is about non-thermal production; the thermal value is quoted there for contrast.
- arXiv:2609.02775 uses "the cosmologically preferred mass, $m_\chi \simeq 1.08$ TeV".
- arXiv:2609.02608 states that the relic abundance "fixes the mass near 1.1 TeV".
- arXiv:2403.19598 states that a thermal higgsino-like state "must be between about 100 and 1150 GeV".

The Planck 2018 abstract (arXiv:1807.06209) gives $\Omega_c h^2 = 0.120 \pm 0.001$.

Only arXiv:2410.15733 states its Sommerfeld and coannihilation treatment, so the cause of the 1.08 versus 1.2 TeV spread was not established. A "1.08–1.2 TeV band with central value near 1.1 TeV" is an inference from the spread of quoted values, not a published result. The assistant also relayed, from abstracts, that lighter thermal higgsinos are under-abundant by a factor of 5–20 in natural SUSY (arXiv:1803.11210).

For the LZ event, the extended-window search (arXiv:2609.02823) reports an exposure of 2.84 t·yr, a window up to about 270 keV, and one event at $248 \pm 23\,(\text{stat}) \pm 23\,(\text{sys})$ keV. Its global significance is 2.6σ and its maximum local significance 3.4σ. In the first pass these quotes came from OpenAlex and were marked contested.

Preferred splittings quoted in the literature are:

- about 350 keV for the Standard Halo Model and about 500 keV with Large Magellanic Cloud effects (arXiv:2609.01504);
- $\delta \simeq 377$ keV (arXiv:2609.02608);
- 481–519 keV for $m_\chi = 1000$–1183 GeV including the LMC tail (arXiv:2609.01590). The same abstract reports that ten years of IceCube data exclude $\delta \lesssim 475$ keV near elastic blind spots;
- a sideband optimum of 339 keV at 1 TeV (arXiv:2609.26698, as relayed by the assistant).

The decisive constraint is the IceCube solar-capture bound, $\delta > 566$ keV at $m_\chi \simeq 1.08$ TeV (arXiv:2609.02775), verbatim cited in the second pass. That abstract states that it "exclude[s] the interpretation of the recent LZ event in terms of endothermic inelastic scattering of Higgsino dark matter". The assistant additionally relayed, without verbatim quotes, the following:

- supporting solar-capture results (arXiv:2609.11833, arXiv:2609.07807, and 0.51–0.56 MeV in arXiv:2609.06760);
- sideband tension (arXiv:2609.09385, arXiv:2609.26698);
- the claim that halo uncertainties cannot close the gap (arXiv:2609.15321);
- a surviving Planck-compatible branch at $\delta \approx 0.85$–1.34 MeV (arXiv:2610.00226).

The abstract of arXiv:2609.09136 states that the thermal higgsino with $\delta \sim 350$ keV is "essentially excluded" and proposes a non-higgsino inelastic candidate.

3.2 Relic density from the tool

Table 1 lists the tool's relic densities. The values rise almost exactly as m^2, as expected from the built-in $\langle\sigma v\rangle \propto m^{-2}$, with $x_f \approx 24.4$–24.8 and $g_* = 86.25$. Linear interpolation crosses 0.120 at ≈ 1158 GeV, with ± 6 GeV for the tool's ± 0.0012 band and about ± 5 GeV for ± 0.001. Direct calls give 0.120 at 1159 GeV and 0.1202 at 1160 GeV.

The tool's "consistent with Planck" flag reads True at every mass from 1000 to 1300 GeV despite tensions of 7–25σ, and False only at 900 GeV (39σ). This appears to be a bug.

With the higgsino flag off at 1100 GeV, $\Omega h^2 = 0.1342$, 0.1199 and 0.0909 for $\langle\sigma v\rangle = 2.0$, 2.25 and 3.0×10^{-26} cm^3/s. Hence $\Omega \propto 1/\langle\sigma v\rangle$ with $\Omega h^2 \langle\sigma v\rangle \approx 0.27 \times 10^{-26}$ cm^3/s, and the Planck abundance requires 2.25×10^{-26} cm^3/s at 1.1 TeV.

Under the built-in scaling, a 10% change in $\langle\sigma v\rangle$ moves the crossing mass by about 5% (55–60 GeV). Matching the literature masses 1.08, 1.10 and 1.20 TeV would require effective factors of

m_χ (GeV)	$\langle\sigma v\rangle$ (10^{-26} cm^3/s)	Ωh^2	tool tension (σ)
900	3.73	0.0731	39.1
1000	3.03	0.0898	25.1
1080	–	0.1045	12.9
1100	2.50	0.1083	9.8
1159	2.25	0.1200	0.0
1160	–	0.1202	0.1
1200	2.10	0.1285	7.1
1300	1.79	0.1503	25.2

Table 1: Tool relic density with the higgsino flag set. The tension is against 0.12 ± 0.0012. Values at 1080 and 1160 GeV come from `check_all_constraints`.

about 0.87, 0.90 and 1.07 on the tool's own rate. These are not physical Sommerfeld factors. The critic estimated that the Planck uncertainty shifts the mass by only $\sim \pm 6$ GeV, negligible compared with the literature spread. The planned with/without-Sommerfeld comparison could not be done because the tool has no such switch.

3.3 Inelastic LZ counts and the kinematic edge

At $m_\chi = 1100$ GeV the predicted LZ count falls steeply with splitting:

- $\delta = 0$: $N_{\rm LZ} = 1.10 \times 10^7$;
- $\delta = 200$ keV: 5.1×10^4;
- $\delta = 300$ keV: 744.6;
- $\delta = 330$ keV: 57.7;
- $\delta = 350$ keV: 1.97, with $P(N \geq 1) = 0.86$ and peak $E_R = 315$ keV;
- $\delta = 355$ keV: 0.494;
- $\delta = 360$ keV: 0.085;
- $\delta = 365$ keV: 0.009;
- $\delta = 370$, 375, 380, 390, 500, 566 and 700 keV: 0. The spectrum peaks would sit at 333–630 keV.

The 248 keV event requires $v_{\rm min}$ of 762, 774, 781, 787 and 799 km/s at $\delta = 350$, 360, 365, 370 and 380 keV. It is therefore accessible below the 776 km/s cutoff only up to $\delta \approx 360$ keV. At $\delta = 380$ keV, $v_{\rm min}(248\,{\rm keV})$ is 802, 799 and 796 km/s for $m = 1000$, 1100 and 1200 GeV, so the edge depends only weakly on mass.

The hand estimate $\delta_{\rm max} = \mu v_{\rm max}^2/2$ gives ≈ 368 keV for 776 km/s and ≈ 385 keV for $544 + 250$ km/s, consistent with the tool's closure between 360 and 370 keV. The zero counts at $\delta \geq 370$ keV are therefore produced by the hard velocity cutoff, not by the energy window. This explains why the first-pass constraint checks at $\delta = 400$ keV returned $N_{\rm LZ} = 0$, a result initially flagged as contradicting the literature.

Log-linear interpolation gives $\delta(N = 1) = 352.5$ keV and $\delta(N = 0.3) = 356.4$ keV at 1100 GeV, so the "about one event" band is only ~ 4 keV wide. In the second pass the window was described as ~ 15 keV wide in the broader sense of counts falling from 57.7 at 330 keV to zero by 370 keV. Along the thermal line the mass dependence is weak:

- at 1159–1160 GeV, $\delta = 360$ keV gives $N = 0.135$–0.136, compared with 0.085 at 1100 GeV;
- at 1160 GeV, $\delta = 350$ keV gives 2.52;
- $\delta(N = 1) \approx 353$ keV at 1160 GeV.

At $\delta = 350$ keV, the checks at 1080, 1100, 1160 and 1200 GeV give $N_{\rm LZ} = 1.79$, 1.97, 2.52 and 2.90. The annual modulation at 1100 GeV, 350 keV and $E_R = 248$ keV has a fractional amplitude of 0.985 peaking on June 2; on September 1 the rate is 22.5% of the peak.

Comparison with the literature fits gives a mixed picture:

- The 350 keV value of arXiv:2609.01504 agrees directly with the tool.
- The 377 keV value of arXiv:2609.02608 lies 24.5 keV above the tool's $\delta(N = 1)$. About 17 keV of that offset would be explained if that paper used a 794 km/s cutoff, which is an assumption. The remaining 5–8 keV is unexplained.
- The 339 keV sideband optimum of arXiv:2609.26698 does not depend on normalization, so it is not a like-for-like comparison. With the tool's normalization such splittings give tens to hundreds of events.

Halo sensitivity could not be computed because the tool has no halo inputs. Analytically, $v_{\rm esc} = 500$ or 600 km/s ($v_{\rm max} \approx 732$ or 832 km/s) would move the edge to ≈ 327 or 423 keV. The first estimate held that 377 keV needs a $v_{\rm max}$ only 1–2% higher ($v_{\rm esc} \gtrsim 555$ km/s). The critic argued that the relevant quantity is $v_{\rm min}$ at 248 keV, which requires $v_{\rm max} \gtrsim 795$ km/s, about 2.5% higher, or $v_{\rm esc} \approx 563$ km/s for an assumed $v_E \approx 232$ km/s. On this basis, bridging 352 and 377 keV by halo uncertainty is plausible but not demonstrated. All halo choices considered stay below the 566 keV solar bound.

3.4 Constraint checker and scans

`check_all_constraints` rated every inelastic point "VIABLE". This includes $\delta = 200$ keV at 1100 GeV, where $N_{\rm LZ} = 5.1 \times 10^4$, so the overall pass flag ignores direct detection.

Only $\delta = 0$ was excluded on all three counts:

- LZ: $N_{\rm LZ} = 1.1 \times 10^7$;
- Fermi-LAT: $\langle\sigma v\rangle_0 = 2.5 \times 10^{-24}$ cm^3/s, against a limit of 2.3×10^{-25} cm^3/s;
- CMB: $p_{\rm ann} = 3.4 \times 10^{-28}$, against a limit of 3.2×10^{-28}.

For $\delta > 0$, the tool's present-day annihilation rate is $\sim 2 \times 10^{-42}$ to 5×10^{-41} cm^3/s, labelled "p-wave + mass-splitting suppressed". The assistant judged this inappropriate for an s-wave, Sommerfeld-enhanced higgsino with $\langle\sigma v\rangle \sim 10^{-26}$ cm^3/s. The first-pass critic marked this finding contested.

The 17×17 scan reported 144 of 289 points viable and 9 predicting 0.3–5 LZ events, with best fit (988 GeV, 350 keV). The tool's own relic curve gives $\Omega h^2 \approx 0.087$ at that best fit, so the point is under-abundant.

The 15×15 scan reported 210 of 225 points viable, 15 predicting 0.3–5 events (apparently one row near 350 keV), and best fit (1000 GeV, 350 keV) on the scan edge. Because of the flag problems, these viability counts are not physical statements.

3.5 Elastic SI placement

LZ 4.2 t·yr (arXiv:2410.17036) states no 1 TeV value in its text. HEPData record 155182 gives, at 1008 GeV, an observed 90% CL limit of 3.056×10^{-47} cm^2 and a median sensitivity of 6.834×10^{-47} cm^2, scaling roughly as $m^{0.985}$. Scaled to 1.1 TeV these become $\approx 3.3 \times 10^{-47}$ (observed) and $\approx 7.5 \times 10^{-47}$ cm^2 (median). The units were not checked against the figure.

XENONnT 3.1 t·yr (arXiv:2502.18005) states $3.7 \times 10^{-46}\,\mathrm{cm}^2 \times (M/1\,\mathrm{TeV})$ above 200 GeV, which gives $\approx 4.1 \times 10^{-46}$ cm^2 at 1.1 TeV. In the first pass, from abstracts only, its minimum limit was 1.7×10^{-47} cm^2 at 30 GeV, and $\sim 10^{-45}$ cm^2 at 1 TeV was only an extrapolation. That earlier estimate was superseded.

For the first runs:

- XENONnT (arXiv:2303.14729) gives $6.08 \times 10^{-47}\,\mathrm{cm}^2 \times (M/100\,\mathrm{GeV})$, about 6.7×10^{-46} cm^2 at 1.1 TeV. That run used a more conservative median power constraint, and the improvement to the current result is about 1.6×.
- LZ minimum limits went from 9.2×10^{-48} cm^2 at 36 GeV (arXiv:2207.03764) to 2.2×10^{-48} cm^2 at 40 GeV, about 4.2×. The first-run LZ value at 1 TeV was not obtained.

The pure-higgsino loop-level cross section is only bounded. Hill and Solon give $\sigma_{\mathrm{SI}} \lesssim 10^{-48}$ cm^2 at 95% CL, independent of M in the heavy-WIMP limit (arXiv:1309.4092 Eq. 5; arXiv:1409.8290). No central value is given because of destructive spin-0/spin-2 interference; for comparison the pure wino is $1.3^{+1.2+0.4}_{-0.5-0.3} \times 10^{-47}$ cm^2 in arXiv:1309.4092. Chen and Hill (arXiv:1912.07795) keep it at or below 10^{-48} cm^2 in the TeV regime. An NLO MSSM study (arXiv:2607.20588) finds cancellations that can push it below 10^{-50} cm^2.

Taking the xenon SI floor as 5.0×10^{-48} cm^2 at 1 TeV, about 5.5×10^{-48} cm^2 at 1.1 TeV (O'Hare, arXiv:2109.03116 data file), the higgsino at 1.1 TeV lies:

- at least $\sim 33\times$ below the LZ observed limit;
- at least $\sim 75\times$ below the LZ median sensitivity;
- at least $\sim 400\times$ below XENONnT;
- at least $\sim 5.5\times$ below the floor, i.e. inside the neutrino fog.

All ratios are lower bounds, because the higgsino value is itself an upper bound. The LZ limit is itself only about 6× above the floor at 1 TeV. XLZD (arXiv:2410.17137) was relayed as reaching 3×10^{-49} cm^2 evidence potential at 40 GeV.

3.6 Gaugino scale implied by the splitting

Using $\delta_0 \simeq m_Z^2(s_W^2/M_1 + c_W^2/M_2)$ (arXiv:1410.4549) with $M_1 = M_2 = M$ and $m_Z^2 = 8315$ GeV^2 gives:

δ (keV)	M (GeV)
350	2.38×10^7
370	2.25×10^7
385	2.16×10^7
566	1.47×10^7

The xenon-accessible window thus needs gauginos at ~ 2.2–2.4×10^4 TeV, about three orders of magnitude above the 10–30 TeV assumed in the initial plan; the critic flagged that error. The solar bound requires $M \lesssim 1.5 \times 10^4$ TeV. With wino exchange only, these values are about 22% lower, and loop and running corrections (arXiv:1410.4549, arXiv:2609.09830) change them by tens of percent. A sub-agent reading of the same literature put the gaugino scale for 100–400 keV at roughly 2–8×10^4 TeV, with Higgs-exchange $\sigma_{\rm SI} \sim 10^{-53}$ cm^2 there.

A tree-level Higgs-exchange estimate, $\sigma_{\rm SI} \sim 5 \times 10^{-46}\,{\rm cm}^2\,(1\,{\rm TeV}/M)^2(1 + {\rm sgn}\,\mu \sin 2\beta)^2$ (from arXiv:2606.31013 and arXiv:1410.4549), reaches the LZ limit at $M \sim 4$ TeV. That scale corresponds to splittings far above the inelastic window. The resulting claim that inelastic and Higgs-exchange elastic signals are mutually exclusive was marked contested by the critic.

3.7 Indirect detection (abstracts only)

The full text of arXiv:2405.13104 could not be downloaded, and all indirect-detection findings were marked contested. As relayed from abstracts:

- A 15-yr Fermi-LAT line-like search comes "within an order one factor" of detecting the higgsino for some Milky Way analog profiles, with H.E.S.S. subleading (arXiv:2405.13104).
- The higgsino is "out of reach by at least a factor of a few" even at the ultimate sensitivity of current H.E.S.S.-like telescopes (arXiv:2210.03140).
- The H.E.S.S. Inner Galaxy Survey (arXiv:2608.07234, 546 h) quotes a line limit of 2.3×10^{-28} cm^3/s at 1 TeV for an Einasto profile and says the thermal higgsino is "probed". The abstract does not state exclusion.
- No MAGIC-specific higgsino limit was found.

For future reach, CTA could be the first experiment sensitive to the thermal higgsino (arXiv:2008.00692). CTA and SWGO could detect the line at high significance for all but the most pessimistic profiles (arXiv:2405.13104). CTAO-North could reach the required sensitivity within a decade (arXiv:2506.08084).

4 Discussion

The combined picture is consistent across the tool and the literature. A thermal higgsino at ~ 1.1 TeV is excluded as a Dirac-like state: Z-mediated elastic scattering overshoots LZ by about seven orders of magnitude. A neutral splitting is therefore mandatory. Below $\delta \approx 330$ keV the inelastic rate is far too large.

Only a narrow band near 350 keV, bounded above by the halo velocity cutoff, produces of order one LZ event. That band lies below the IceCube solar-capture bound $\delta > 566$ keV of arXiv:2609.02775. Splittings allowed by solar capture are beyond xenon's inelastic reach under any halo considered. In that regime only loop-level elastic scattering remains, at $\lesssim 10^{-48}$ cm^2, which is inside the neutrino fog and at least $\sim 33\times$ below current LZ sensitivity. The higgsino is therefore not excluded by direct detection, but current xenon experiments essentially cannot see it unless gaugino mixing raises the elastic rate.

The tool's quantitative content is limited. Its relic curve is a calibrated scaling with no Sommerfeld, coannihilation or bound-state physics, so the 1159 GeV crossing measures the calibration rather than higgsino physics. The `is_higgsino` flag silently ignores a supplied $\langle\sigma v\rangle$. The Planck and overall-viability flags are unreliable, and the indirect-detection model is physically inappropriate for a higgsino. The halo is hard-coded, so the window's dominant uncertainty

could not be tested. The tool also contains no published LZ limit curve to cross-check against. Its self-description attributes arXiv:2609.02608 to the LZ Collaboration, whereas metadata identifies that paper as a single-author interpretation by Di Mauro.

The literature side carries its own limitations. In the first pass, every arXiv-backed tool failed, so all citations rested on OpenAlex abstracts. In the second pass verbatim abstract citations succeeded, but full texts were never downloaded. The LZ 1 TeV limit came from HEPData rather than the paper text, and the `cite` quote check on arXiv:2410.17036 failed. Indirect-detection statements remain at abstract level, and the 1.08–1.2 TeV mass spread was not explained. The assistant's summary text also twice incorrectly claimed that no tool calculations had been performed.

5 Conclusions

Established in the session.

- The cited literature places the thermal pure higgsino at about 1.08–1.2 TeV. Planck gives $\Omega_c h^2 = 0.120 \pm 0.001$.
- The tool's calibrated relic curve gives $\Omega h^2 = 0.0731$–0.1503 over 0.9–1.3 TeV and crosses Planck at ≈ 1159 GeV.
- In the tool, elastic Z exchange is excluded. About one LZ event requires $\delta \approx 352$ keV at a fixed 776 km/s cutoff, with closure by 370 keV. The literature solar-capture bound $\delta > 566$ keV lies beyond that reach.
- The loop-level elastic cross section, $\lesssim 10^{-48}$ cm^2, lies at least $\sim 33\times$ below LZ, $\sim 400\times$ below XENONnT, and inside the neutrino fog at 1.1 TeV.
- The xenon-accessible window corresponds to gauginos at $\sim 2 \times 10^4$ TeV.

Contested or unresolved.

- The mutual exclusivity of the inelastic and Higgs-exchange elastic signals.
- The indirect-detection status, which is "not excluded, profile-dependent, CTAO decisive" at abstract level only

Data and code availability

All outputs of this session are collected in the project folder `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-higgsino-dd`. Full documentation of how these results were obtained is in the supplementary material (`supplement.pdf`: implementation details, the tool-call log, the research board with every finding and its evidence, figures not shown here, and the code index) and the verbatim transcript (`transcript.pdf`), generated alongside this report.